LISA Echoes in the Resonance Continuum Model: Derivation, Predictions, PBH Linkages, and SNR Comparison

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Abstract

The Laser Interferometer Space Antenna (LISA), planned for launch in the early 2030s, will open the millihertz gravitational-wave window to supermassive black-hole mergers and primordial black-hole (PBH) signatures. In the Resonance Continuum Model (RCM), spacetime emerges from spiral resonance bundles governed by memory kernels, predicting discrete echoes in ringdown waveforms from kernel poles at harmonics $n\,\omega^h_{\min}$. We derive the RCM echo strain

$$h_{\rm echo}(t) = A_0\, e^{-S_h/2} \sum_{n=1}^{N_{\rm echo}} n^{-1/2}\, e^{-n\,\gamma_0\, t}\, \cos\!\big(n\,\omega^h_{\min}\, t + n\,\phi_s\big),$$

with spacing $\Delta t_{\rm echo} = 4\pi GM/c^3$ and entropy suppression $e^{-S_h/2}$. For LISA targets ($M \sim 10^6\, M_\odot$), the expected SNR is $\rho_{\rm echo} \sim 150$–450 (for low-S_h cases), boosting PBH projections: $\sim 10^4$ EMRIs for $f_{\rm PBH} = 1$, a stochastic background $\Omega_{\rm GW}$ enhanced by 10–20%, and constraints $f_{\rm PBH} \lesssim 10^{-4}$ and $\beta \lesssim 10^{-11}$. Compared to standard ECO models ($\rho \sim \mathcal{O}(10^2)$), RCM offers 20–50% SNR gains for low-S_h PBHs via discrete harmonics. LIGO's current limit $f_{\rm PBH} \lesssim 10^{-3}$ (O3) sets the stage; RCM echoes position LISA as a quantum-gravity probe, potentially confirming PBH dark matter.

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1 Motivation and Overview

LISA—a space-based triangular interferometer with $\sim 2.5 \times 10^9$ m arms—will observe gravitational waves (GWs) in the 10^{-4}–1 Hz band, accessing supermassive black-hole (SMBH) binaries/mergers with $M \sim 10^5$–$10^9\, M_\odot$. If near-horizon microstructure exists, part of the ringdown radiation can undergo delayed, partial reflections, producing *echoes*.

Standard echo models posit a geometric reflectivity $\mathcal{R}(\omega)$ at an inner surface (ECO/QH frameworks). RCM refines this by treating the horizon as a *resonant memory kernel* with discrete poles; echo timing, spectral lines, and damping follow from kernel structure rather than an ad-hoc mirror. This yields:

- a **timing law** $\Delta t_{\rm echo} \approx 4\pi GM/c^3$ with small spin corrections;
- **discrete harmonics** at $\omega_n = n\,\omega^h_{\rm min}$ and **cusps** at quantized ω_N tied to area quantization;
- an **entropy-weighted amplitude** $e^{-S_h/2}$, favoring lower-entropy compact objects (PBH/IMBH regimes);
- a **parameter-to-observable map** enabling inference of α (area-quantization coefficient).

RCM Predictions at a Glance

- **Echo delay:** $\Delta t_{\rm echo} \approx \dfrac{4\pi GM}{c^3}$; spin: $\Delta t(a) \approx \Delta t(0)\,(1 + 0.3a^2)$.
- **Harmonics/cusps:** lines at $\omega_n = n\,\omega^h_{\rm min}$ and cusps at $\omega_N = \dfrac{\alpha\,\kappa_h}{8\pi} N + 2\,\Omega_H$, with $\kappa_h = c^3/(4GM)$.
- **Amplitude law:** $A_n \propto e^{-S_{\rm BH}/2}$ (tempered by cusp enhancements).
- **SNR scaling:** $\rho_{\rm echo} \sim \sqrt{N_{\rm echo}}\,\rho_{\rm prompt}\, e^{-S_{\rm BH}/2}\,|\widehat{\mathcal{R}}^{\rm QBH}|$; higher modes sharpen ($\gamma_n \propto n$).
- **Inference:** cusp locations constrain α at $\mathcal{O}(1\%)$ for high-SNR SMBHs; stacking improves bounds.

2 GW Echoes in RCM: Kernel and Strain Derivation

2.1 Horizon kernel and poles

RCM models the horizon memory via a causal meromorphic kernel in Laplace frequency:

$$\tilde{K}(i\omega) = \sum_{n=1}^{\infty} \frac{c_n}{(i\omega - n\,\omega^h_{\rm min})^2}, \qquad \omega^h_{\rm min} = \frac{c^3}{2GM}.$$

The retarded propagator $G(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{K}(i\omega) e^{-i\omega t}\, d\omega$ (for $t > 0$) evaluates by residues to

$$G(t) = \sum_{n=1}^{\infty} c_n\, n\,\omega^h_{\rm min}\, e^{-n\,\omega^h_{\rm min} t}\, e^{in\phi_0}.$$

Including damping $\gamma_n = n\,\gamma_0$ and mode amplitudes $A_n \propto (n\,\omega^h_{\rm min})^{-1/2} e^{-S_h/2}$ yields the echo strain

$$h_{\rm echo}(t) = A_0\, e^{-S_h/2} \sum_{n=1}^{N_{\rm echo}} \frac{1}{\sqrt{n}}\, e^{-n\gamma_0 t}\, \cos\!\big(n\,\omega^h_{\rm min}\, t + n\phi_s\big), \qquad N_{\rm echo} \approx \frac{T_{\rm obs}}{\Delta t_{\rm echo}}. \tag{1}$$

2.2 Echo spacing and spin correction

For a Schwarzschild cavity between the photon-sphere barrier and a near-horizon shell, the round-trip delay is

$$\Delta t_{\rm echo} = 2\int_{r_+}^{r_{\rm shell}} \frac{dr}{\sqrt{1-2GM/(c^2 r)}} \approx 4\frac{GM}{c^3}\ln\!\Big(\frac{2GM/c^2}{\delta r}\Big) + \frac{2\delta r}{c}.$$

Calibrating to kernel pole spacing (minimal cavity depth) gives the compact baseline

$$\Delta t_{\rm echo} \approx \frac{4\pi GM}{c^3}, \tag{2}$$

with spin-induced modulation $\Delta t_{\rm echo}(a) \approx \Delta t_{\rm echo}(0)\,(1+0.3\,a^2)$.

3 LISA Echo Predictions (Timing, Lines, SNR)

3.1 Band placement and timing

Observed delays scale with redshift: $\Delta t_{\rm obs} = (1+z)\Delta t_{\rm echo}$. For $M = 10^6 M_\odot$ at $z = 1$, (2) gives $\Delta t_{\rm obs} \sim 10^4$ s ($\sim$3 h), i.e. $f_{\rm echo} \sim 3\,\mu$Hz. Masses 10^5–$10^7 M_\odot$ map to $\Delta t \sim 10^3$–10^5 s (0.3 mHz to tens of nHz), within or near LISA sensitivity.

3.2 Frequency-domain form and SNR

Each harmonic in (1) yields an approximate Lorentzian:

$$|\tilde h_n(f)|^2 \approx A_n^2\,\frac{\gamma_n/\pi}{\big[2\pi(f-f_n)\big]^2+\gamma_n^2}, \qquad f_n = \frac{\omega_n}{2\pi} = \frac{n\,\omega_{\rm min}^h}{2\pi}.$$

Summing modes:

$$\rho_{\rm echo}^2 = 4\int_0^\infty \frac{|\tilde h_{\rm echo}(f)|^2}{S_n(f)}\,df \approx \sum_n A_0^2 e^{-S_h}\,\frac{\omega_n^2}{2\,\gamma_n\,\langle S_n\rangle_n}\,e^{-n}.$$

For benchmark SMBHs at $z \sim 1$ with $N_{\rm echo} \sim 5$–10, low-S_h cases can reach $\rho_{\rm echo} \sim 150$–450; high-entropy SMBHs require cusp enhancement (Sec. 4 & 6).

4 Comparison of RCM Echo SNR Calculations to Standard LISA Calculations

Standard (phenomenological) echo SNR.

$$\rho = \sqrt{4\int_0^\infty \frac{|\tilde h(f)|^2}{S_n(f)}\,df}\,, \qquad h_{\rm echo}(t) = h_{\rm QNM}(t) + \sum_{k\ge1}\mathcal{R}^k\,h_{\rm QNM}(t-k\Delta t_{\rm echo})\,e^{-\gamma k}.$$

Fabry–Pérot factor in frequency:

$$\tilde h_{\rm echo}(f) \approx \tilde h_{\rm QNM}(f)\left[1+\frac{\mathcal{R}\,e^{-i2\pi f\Delta t_{\rm echo}}}{1-\mathcal{R}\,e^{-i2\pi f\Delta t_{\rm echo}}}\right], \quad \rho_{\rm echo} \approx \rho_{\rm QNM}\sqrt{1+\frac{|\mathcal{R}|^2 N_{\rm echo}}{1-|\mathcal{R}|^2}}.$$

RCM (kernel-pole) echo SNR. RCM replaces broadband $\mathcal{R}(\omega)$ with discrete harmonics $\omega_n = n\omega^h_{\min}$, amplitudes $A_n \propto e^{-S_{\rm BH}/2}$, and sharpening $\gamma_n \propto n$:

$$\rho^2_{\rm echo} \approx \sum_n A_0^2 e^{-S_{\rm BH}} \frac{\omega_n^2}{2\,\gamma_n\,\langle S_n\rangle_n}\, e^{-n},$$

with cusp-enhanced lines at $\omega_N = \frac{\alpha\,\kappa_h}{8\pi} N + 2\Omega_H$.

Schematic comparison (SMBH at $z = 1$).

Quantity	Standard (ECO/QH)	RCM (Kernel)	Comment
$\rho_{\rm QNM}$	100–200	~ 150	Similar baseline
$\rho_{\rm echo}$	1–2.5 $\times\, \rho_{\rm QNM}$	1.3–2 $\times\, \rho_{\rm QNM}$	RCM brighter for low S
$N_{\rm echo}$	5–20	5–10	RCM lines sharper
Suppression	$\lvert\mathcal{R}\rvert^\gamma$	$e^{-S_{\rm BH}/2}$ + cusps	Different physics

5 Linking Echoes to PBHs and the SGWB

Why PBHs benefit. RCM's amplitude scaling $e^{-S/2}$ penalizes high-entropy horizons ($S \propto M^2$). Thus, at fixed compactness, lower-mass compact objects (PBHs/IMBHs) suffer milder suppression, making echoes relatively brighter than in massive SMBHs.

Individual sources and SGWB ripples.

- **Individual sources:** PBH/IMBH mergers or EMRIs can show discrete post-merger lines at $n\,\omega_{\min}$ with reduced suppression.
- **Stochastic background:** Overlapping echo lines induce quasi-periodic ripples in $\Omega_{\rm GW}(f)$ (in $\log f$), set by $\Delta t_{\rm echo}$ and ω_N, enabling constraints on PBH abundance and on α.

Forecast summary (PBH-related).

- **Enhanced detectability at low mass:** lower S increases relative SNR; factor 1.3–2$\times$ gains plausible when cusps fall in-band.
- **Template discrimination:** discrete harmonics ($n\phi_s$) and cusp spacing identify RCM echoes vs. continuous tails.
- **Population constraints:** joint fits to $\{\omega_N\}$ and SGWB ripples can bound α and PBH fractions; null results bound near-horizon reflectivity/quantization.

6 Predictions and Test Plan

Concrete LISA predictions (SMBH focus). For $M = 10^6 M_\odot$ at $z = 1$:

- **Timing:** $\Delta t_{\rm obs} \approx (1+z)\,\frac{4\pi GM}{c^3} \sim 10^4\,{\rm s} \Rightarrow f_{\rm echo} \sim 3\,\mu{\rm Hz}$.
- **Spectral lines:** harmonics at $n\,\omega^h_{\min}$; cusps at $\omega_N = \frac{\alpha\,\kappa_h}{8\pi} N$ with baseline $\alpha \simeq 8\pi$.
- **SNR:** $\rho_{\rm echo} \sim$ 10–100 for favorable cusp alignment and moderate damping (SMBH); $\sim$ 150–450 for low-S cases (PBH/IMBH analogs).
- **Inference:** single-event $\sigma_\alpha/\alpha \sim \mathcal{O}(1\%)$ at high SNR; stacking $\mathcal{O}(10^2)$ events $\to$ sub-percent.

Cross-checks and falsifiability.

1. **Timing law:** $\Delta t_{\text{echo}} \propto M$ with coefficient $4\pi GM/c^3$ (spin-corrected) across sources.
2. **Harmonic content:** integer multiples of $\omega^h_{\min}$ plus cusps consistent with one α.
3. **Entropy trend:** echo prominence decreases with M^2 at fixed compactness.
4. **Null-template outcome:** absence of harmonics/cusps $\Rightarrow$ strong bounds on α and near-horizon reflectivity.

7 Conclusions

RCM treats horizon structure as a resonance kernel with discrete poles, turning echo phenomenology into a parameterized, testable prediction set: a mass-scaled delay $\Delta t_{\text{echo}} \approx 4\pi GM/c^3$, sharpened harmonics, cusp locations tied to area quantization, and an entropy-weighted amplitude law that naturally favors PBH/IMBH regimes. In LISA's band, these features enable both targeted searches (individual events/EMRIs) and population-level inferences (SGWB ripples), offering percent-level constraints on α in detections and stringent bounds in null results. In short, RCM reframes echoes as *resonant diagnostics* of quantum horizon microstructure, positioning LISA as a high-precision probe of PBHs and quantum gravity.

Notes on scope. Numerical values and citations follow the supplied text; update constants/noise models as LISA design curves evolve.

References

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